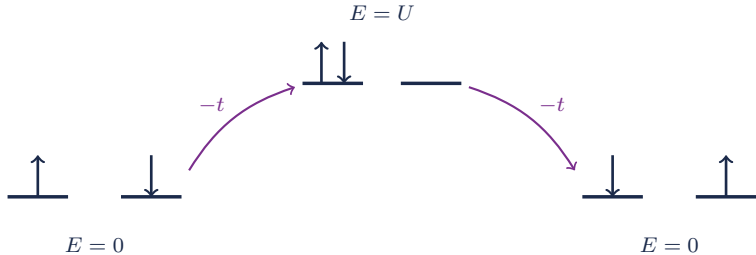


(a) parallel spins:
hop Pauli blocked, $E_{\text{triplet}} = 0$



(b) antiparallel spins: virtual double occupancy at energy U ,
second order gives $E_{\text{singlet}} \approx -4t^2/U$